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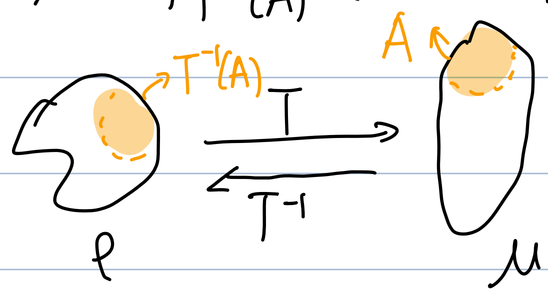
- Optimal Transport problem (Monge Formulation)

  Given two densities of mass (probability measures), $\rho, \mu \geq 0$ on $\mathbb{R}^d$

  $\int \rho(x)\,dx = \int \mu(y)\,dy = 1$,

  find a map $T: \mathbb{R}^d \to \mathbb{R}^d$ s.t. $T\#\rho = \mu$ (push forward constraint)

  ie. $\int_A \mu(y)\,dy = \int_{T^{-1}(A)} \rho(x)\,dx$ for any Borel subset $A \subset \mathbb{R}^d$



  and minimizing transportation cost

  $$M(T) := \int_{\mathbb{R}^d} c(x, T(x))\,\rho(x)\,dx$$

  where $c(x,y)$ is some "cost"

  - Eg. $c(x,y) = \frac{1}{2}\|x-y\|^2$

- Density Estimation (using push forward constraint)
  - measure ρ on X, measure μ on Y
  - $T\#\rho \Rightarrow \mu(A) = (T\#\rho)(A) = \rho(T^{-1}(A))$

    require $T\#\rho = \mu$: $\mathbb{E}_{T\#\rho}[\phi] = \mathbb{E}_\rho[\phi \circ T] = \mathbb{E}_\mu[\phi]$, $\forall$ measurable func. ϕ

    $$\int (\phi \circ T)(x)\,\rho(x)\,dx = \int \phi(y)\,\mu(y)\,dy$$

$$\int \phi(T(x))\, p(x)\, dx = \int \phi(y)\, \mu(y)\, dy$$

Change of variable $(y = T(x))$

$$= \int \phi(T(x))\, \mu(T(x))\, \underbrace{|T'(x)|}_{\text{abs val of det of Jacobian matrix } T(x)}\, dx$$

$\Downarrow$

- $p(x) = \mu(T(x))\, |T'(x)|$

$y = T(x)$
$x = T^{-1}(y)$ $\Rightarrow$ $p(T^{-1}(y)) = \mu(y) \dfrac{1}{|T^{-1\prime}(y)|}$

$\downarrow$

Jacobian of T^{-1}

Jacobian matrix:

$X = [x_1, \dots, x_d]^T \in \mathbb{R}^d$

$T : \mathbb{R}^d \to \mathbb{R}^d = \begin{pmatrix} T_1(x_1, \dots, x_d) \\ \vdots \\ T_d(x_1, \dots, x_d) \end{pmatrix}$

$$T'(x) = \begin{bmatrix} \partial T_1/\partial x_1 & \partial T_1/\partial x_2 & \cdots & \partial T_1/\partial x_2 \\ \vdots & & & \\ \frac{\partial T_d}{\partial x_1} & & \cdots & \frac{\partial T_d}{\partial x_d} \end{bmatrix}$$

- Monte Carlo Integration

a R.V. $X \sim p(x)$ PDF (density), $X_1, \dots, X_N$ iid samples $\sim p$

iid: identically independent distributed

- $\mathbb{E}[\phi] = \mathbb{E}_p[\phi(X)] = \int \phi(x)\, p(x)\, dx$

$$\approx \frac{1}{N} \sum_{i=1}^{N} \phi(X_i) \quad \text{(sample average)}$$

(CLT, Law of Large # $\Rightarrow$ convergence theorem)

- Importance Sampling : $X_1, \dots, X_N \sim \tilde{p}$ (referenced density)

$$\frac{p}{\tilde{p}}(x) = \frac{p(x)}{\tilde{p}(x)} \text{ known}$$

$$\mathbb{E}_p[\phi(X)] = \int \phi(x) \frac{p(x)}{\tilde{p}(x)} \cdot \tilde{p}(x)\, dx = \mathbb{E}_{\tilde{p}}\left[\phi(X) \cdot \frac{p}{\tilde{p}}(x)\right]$$

(weighted sum) $\approx \frac{1}{N}\sum_{i=1}^{N} \varnothing(x_i)\cdot\frac{p}{\tilde{p}}(x_i)$

weight

- Gradient Flow / Gradient Descent
  - ODE

$x(t)$ ← time; ↑ position/displacement

(velocity) $\dot{x} = \frac{dx}{dt} = V(x(t))$

$$x(t) = x(t_0) + \int_{t_0}^{t} V(x(s))\,ds$$

– Eg. $V(x) = V$ const. $\Rightarrow x(t) = x(t_0) + V(t - t_0)$

  - Gradient Flow

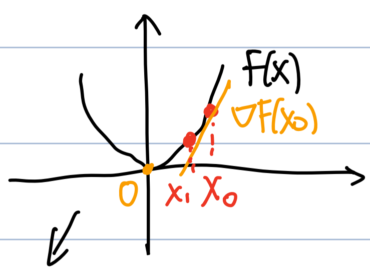
some objective to minimize

$$\left(\frac{x(t_{i+1}) - x(t_i)}{\Delta t} \approx\right) \dot{x} = \frac{dx}{dt} = -\nabla_x F(x(t)) = -\nabla_x F(x)$$

$$\Rightarrow \dot{x} = 0 \text{ (velocity} = 0) \Rightarrow \nabla_x F = 0$$

  - Gradient Descent (discretize flow)

$F(x) = x^2$



F is convex!

$$x^{t+1} = x^t - \eta \cdot \nabla F(x^t)$$

η ↳ learning rate

$$-\nabla F(x) = -2x$$

· Normalizing Flow

$$\frac{dT(x,t)}{dt} = -\frac{\delta KL(\phi \circ T(\cdot,t))}{\delta \phi}\bigg|_{\phi = id}$$

$$T(\cdot, t^n) = \phi^n \circ \phi^{n-1} \circ \cdots \circ \phi^1$$

$$\phi^k = \phi(\cdot, \theta^k) \quad , \text{ assume } \phi(\cdot, \theta_0) = id.$$

$$KL(T(\cdot,t)) = \int \log\Big(\underbrace{\frac{\rho(x)}{\tilde{\rho}_t(x)}}\Big)\rho(x)\,dx \quad (\text{"difference btw } \rho \,\&\, \tilde{\rho}_t\text{"})$$

$$\to T(\cdot,t)\#\rho \sim \tilde{\rho}_t$$

$$KL_t = KL(\underbrace{\phi \circ T(\cdot,t)})$$

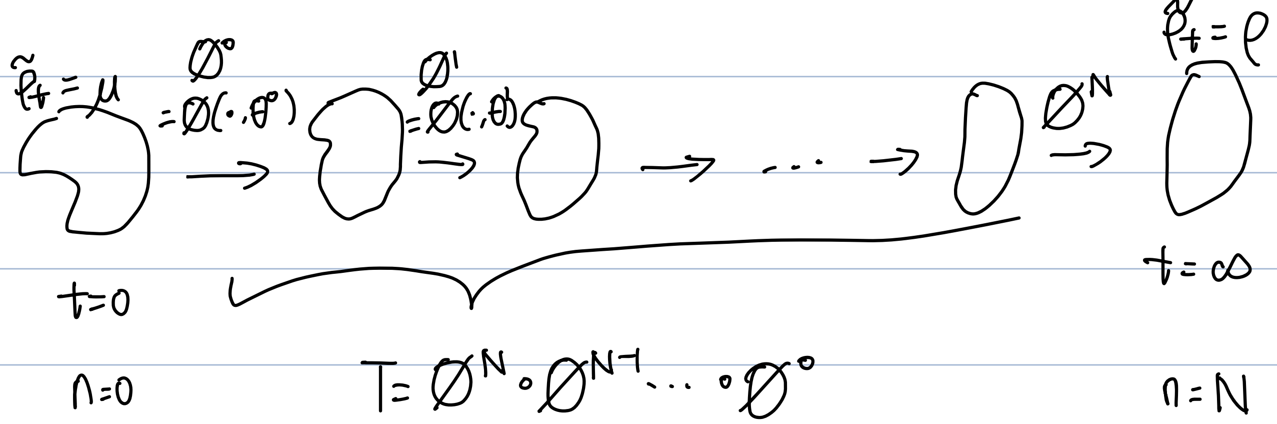
$$\tilde{\rho}_t^{\phi} \sim \phi \circ T(\cdot,t)\#\rho$$

$$\phi(T(\cdot,t), \theta(t)) \sim \tilde{\rho}_t^{\phi}$$

$$\dot{\theta}_t = -\frac{\partial KL_t}{\partial \theta_t}$$

$$\theta^{n+1} = \theta_0 - \frac{\partial KL_t}{\partial \theta_t}\bigg|_{\theta^n}$$

$$\downarrow \quad \phi^{n+1} = \phi(\cdot, \theta^{n+1})$$



$$T = \phi^N \circ \phi^{N-1} \cdots \circ \phi^0$$